

AlphaCondor

Autonomous AI Options Trading System on Alpaca

Combining Google Gemini Reasoning with Deterministic Risk Guardrails
Dual-Engine: Institutional Wheel Strategy + High-Efficiency ODTE Iron Condors



Alpaca AI Trading Agents Hackathon (lablab.ai) | Track 04: Income & Portfolio Overlay Agents

Alpaca Paper Account ID: PA3QPTGUKZ8P (\$100,000 Starting Balance) • GitHub: github.com/prolegendluv/alphacondor

The Options Dilemma & The AlphaCondor Solution

Bridging Generative AI and Quantitative Risk Management

The Retail Options Problem

- Massive Capital Lockup: Selling Cash-Secured Puts on blue chips requires \$30K–\$60K in collateral per contract for small nominal yields.
- Unconstrained LLM Risk: Directly connecting an LLM to order submission invites hallucinations, illegal strikes, and blown accounts.
- Rapid Intraday Gamma: ODTE options decay exponentially but carry tail risk if not actively monitored and closed before the bell.
- Fragmented Tooling: Retail traders lack institutional multi-leg execution and continuous audit logging.

The AlphaCondor Architecture

- Strict AI/Risk Decoupling: Google Gemini acts solely as an analyst and advisor. A deterministic Python firewall validates every trade.
- Dual-Strategy Engine: Combines conservative institutional Wheel income with defined-risk, high-margin ODTE Iron Condors (\$500 margin).
- Deterministic Guardrails: 7 hard mathematical gates enforce cash collateral, concentration limits, and covered share backing.
- Institutional Observability: Real-time WebSocket execution, full SQLite audit journal, Rich CLI, and a live Streamlit dashboard.

System Architecture: End-to-End Pipeline

From Market Perception to Autonomous Broker Execution

1. Perception

Alpaca Trading API

- Stock Bars & Prices
- Option Chains with Greeks
- Alpaca NewsClient

2. Analysis Engine

Quantitative Layer

- Wilder's RSI(14), EMAs
- Black-Scholes IV Rank
- Liquidity & Spread Filters

3. AI Cognition

Google Gemini 3.5

- Context Synthesis
- Strict Pydantic JSON
- Fallback Rule Engine

4. Risk Firewall

7 Deterministic Gates

- Collateral Verification
- Share Backing Check
- Concentration Caps

5. Execution

Alpaca Broker

- Limit Orders (Intent)
 - Multi-Leg Condors
 - TradingStream
- WebSockets

Dual-Strategy Options Engine

Balancing Institutional Stability with Maximum Capital Efficiency



Strategy 1: The Core Wheel

- Asset Focus: Blue-chips & broad indices (AAPL, MSFT, NVDA, SOFI, SPY).
- Phase 1 — Cash-Secured Put (CSP): Sells OTM puts ($\Delta \approx 0.25$, 30–45 DTE) on fundamentally sound assets.
- Phase 2 — Assignment: If assigned, purchases 100 shares at strike, resetting cost basis (Strike - Premium).
- Phase 3 — Covered Call (CC): Sells OTM calls against shares to harvest premium while awaiting exit.
- Target Return: Conservative 20%–35% annualized options income.



Strategy 2: ODTE Iron Condors

- Asset Focus: High-liquidity index ETF (SPY).
- Defined-Risk Structure: 4-leg credit spread (Buy Put + Sell Put + Sell Call + Buy Call).
- 10x–20x Margin Efficiency: Required margin is capped at spread width (\$500/spread) vs \$30K+ for CSPs.
- Intraday Theta Harvest: Exploits non-linear same-day expiration decay.
- Backtest Demonstration: Up to 21x higher 4-day P&L potential (\$2,400–\$7,500 vs \$350) with identical \$100K capital.

AI Cognition & Deterministic Risk Gates

Structured LLM Reasoning Governed by Pure Mathematical Guardrails

Risk Gate	Target Metric	Mathematical Enforcement
1. Confidence Floor	Model Certainty	Hard rejection if Gemini confidence score < 70%.
2. Concurrency Cap	Portfolio Exposure	Restricts maximum concurrent open positions to 20.
3. Cash Collateral	Cash-Secured Check	Requires 100% cash backing for puts: $\text{Strike} \times 100 \times \text{Qty} \leq \text{Available Cash}$.
4. Concentration Cap	Single-Ticker Limit	Capital allocation per underlying cannot exceed maximum threshold.
5. Share Backing	Covered Call Rule	Requires ≥ 100 shares owned. Naked call execution is mathematically impossible.
6. Phase Consistency	State Machine Rule	Enforces valid wheel transitions (cannot sell call without shares; cannot duplicate puts).
7. Liquidity / Spread	Execution Quality	Rejects contracts with bid-ask spread > 15% or prices $\leq \$0.05$.

Intelligent Position Lifecycle Management

Automated Profit Takes, Gamma Rolls, and Stop Losses

50% Profit Target

Locking in Gains Early

Closes open option positions once 50% of maximum potential profit is captured, dramatically accelerating capital velocity and reducing exposure duration.

21 DTE Roll Threshold

Eliminating Gamma Risk

Options experience severe non-linear delta and gamma fluctuations during the final 3 weeks. Positions reaching 21 DTE are automatically rolled out or exited.

200% Stop Loss

Capping Tail Losses

If market volatility pushes a position into a loss exceeding 200% of received premium, the position is immediately liquidated to protect capital.

3:45 PM EOD Exit

Zero Overnight Risk

For ODTE Iron Condors on SPY, any open spreads are automatically closed 15 minutes before the 4:00 PM ET close to avoid surprise assignment.

Alpaca Developer Stack & MCP Server

Leveraging the Full Breadth of Alpaca's Programmable Brokerage

Alpaca Trading API (alpaca-py)

Order placement using explicit PositionIntent (SELL_TO_OPEN, BUY_TO_CLOSE). Account state and buying power synchronization.

Alpaca Market Data API

Real-time quotes, historical OHLCV daily bars, and option chains with Black-Scholes Greeks.

Alpaca TradingStream (WebSocket)

Event-driven asynchronous fill monitor tracking trade updates, cancellations, and order fills.

Alpaca Model Context Protocol (MCP)

Configured via mcp_config.json running uvx alpaca-mcp-server, enabling Claude & Cursor to manage trades conversationally.

Streamlit Web Dashboard

Real-time portfolio KPI monitoring, open position tables, trade logs, and historical equity curve.

Rich CLI & SQLite Audit Trail

Terminal command interface with comprehensive SQLite trade journal recording all contexts, signals, and risk gate verdicts.

Verification, Testing & P&L Analysis

Rigorously Tested Across Unit Tests, Simulations, and Paper Trading

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Unit Tests Passing

Complete pytest coverage of IV Rank, strike selection, Wheel states, and ODTE Condors.

\$100,000

Fresh Account Balance

Pristine Alpaca paper account PA3QPTGUKZ8P ready for competition judging.

7x–21x

Margin Efficiency Multiplier

Demonstrated P&L expansion on SPY credit spreads vs cash-locked single puts.



Key Takeaways for Hackathon Judges

- ✓ Track 04 Fit: Perfectly implements institutional options income (Wheel + Defined-Risk Credit Spreads).
- ✓ Technical Depth: Avoids brittle dependencies, utilizing standard scipy Black-Scholes math and pure pandas indicators.
- ✓ Production Safety: The strict decoupling ensures an LLM can never over-leverage or execute illegal naked calls.
- ✓ Complete Deliverables: Public GitHub repo, Streamlit web app, technical write-up, and dedicated \$100K Alpaca paper account.